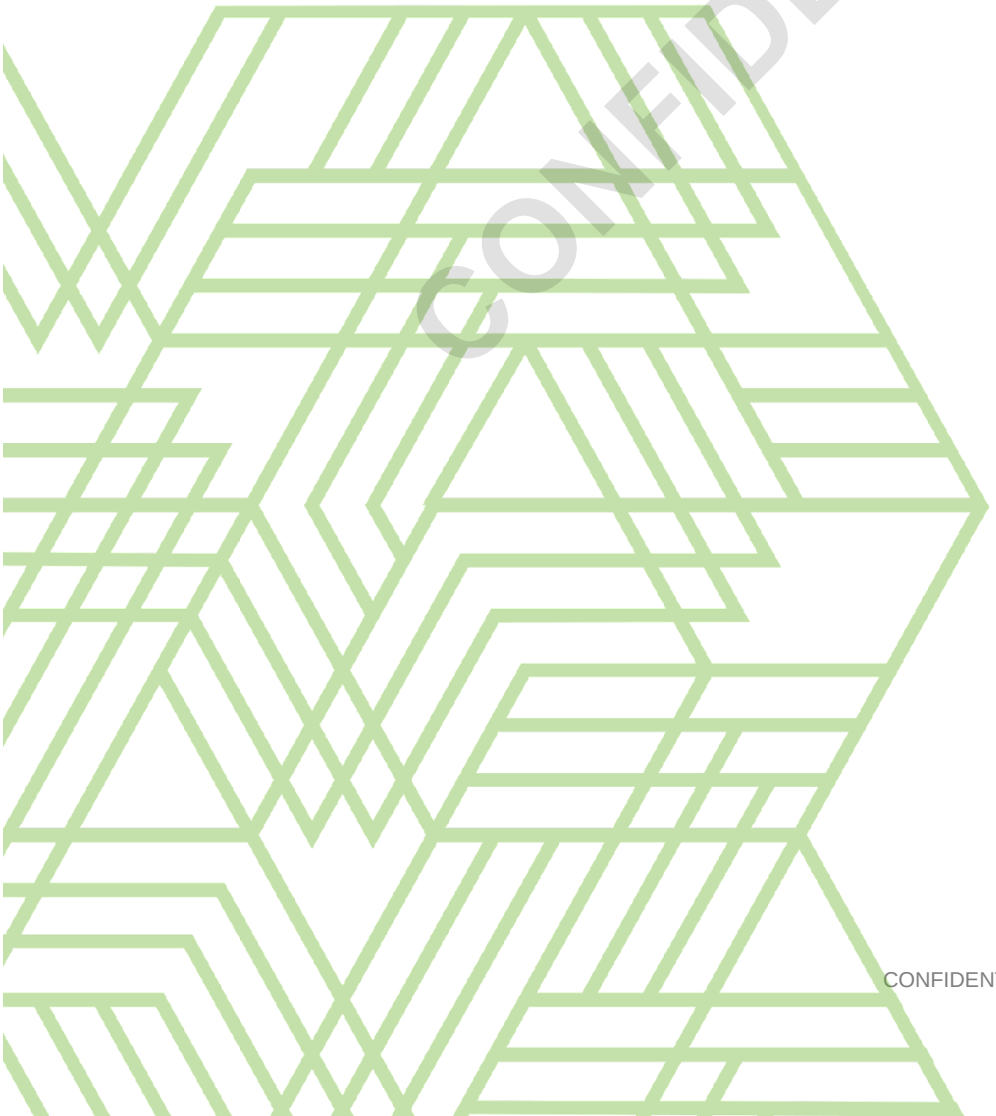
ATM2/ATM3 Series HID Keyboard with Zephyr RTOS

Application Note

SUMMARY: This document provides the functionality, state machine and code flow of the Atmosic HID keyboard which was built with Zephyr-RTOS running on an Atmosic ATM2/ATM3 Wireless SoC.

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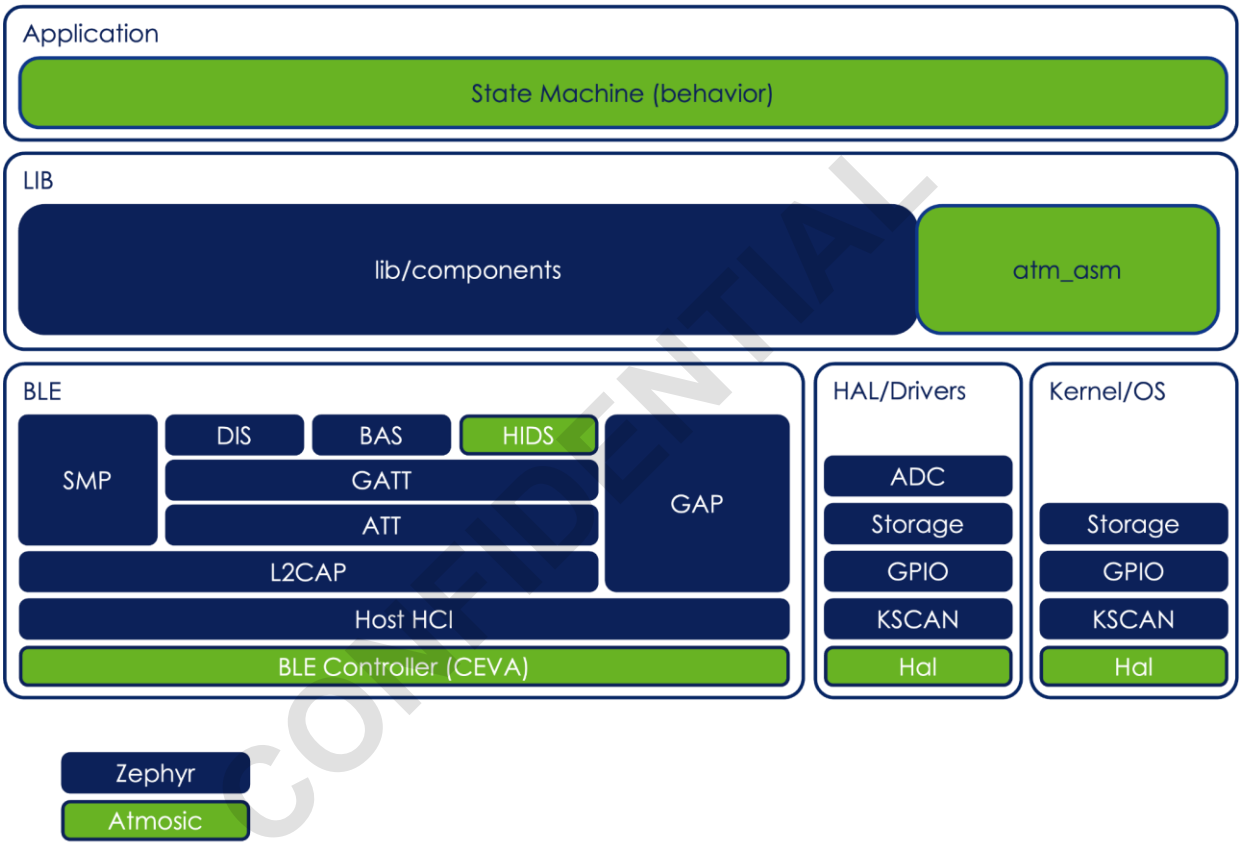
Table 4 - kbd_gap Operations

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1. Overview

This document provides the functionality, state machine and code flow of the Atmosic HID keyboard Reference Design with Zephyr-RTOS running on an Atmosic ATM2/ATM3 Wireless SoC.

1.1 Hierarchical and Components



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Figure 1 - Components Architecture

1.2 Folder Structure



2. Hardware Setup

- The whole keyboard sample includes the following.
  - PV Keyboard (ATMKBD-3221-PV)
    - ATM3221 module (ATM3-MOD-C(6x6)_V3a)
    - Carrier board (ATM3_PV-KEYBOARD_V2)
  - I/O Adaptor board (ATMIAB-01) for FFC connector and FFC cable.
  - Interface board (ATMIOB-Mx2xx-02)
- The set up can be found in Figure 2. Please make sure connection of the FFC cable is the same as shown in Figure 2.

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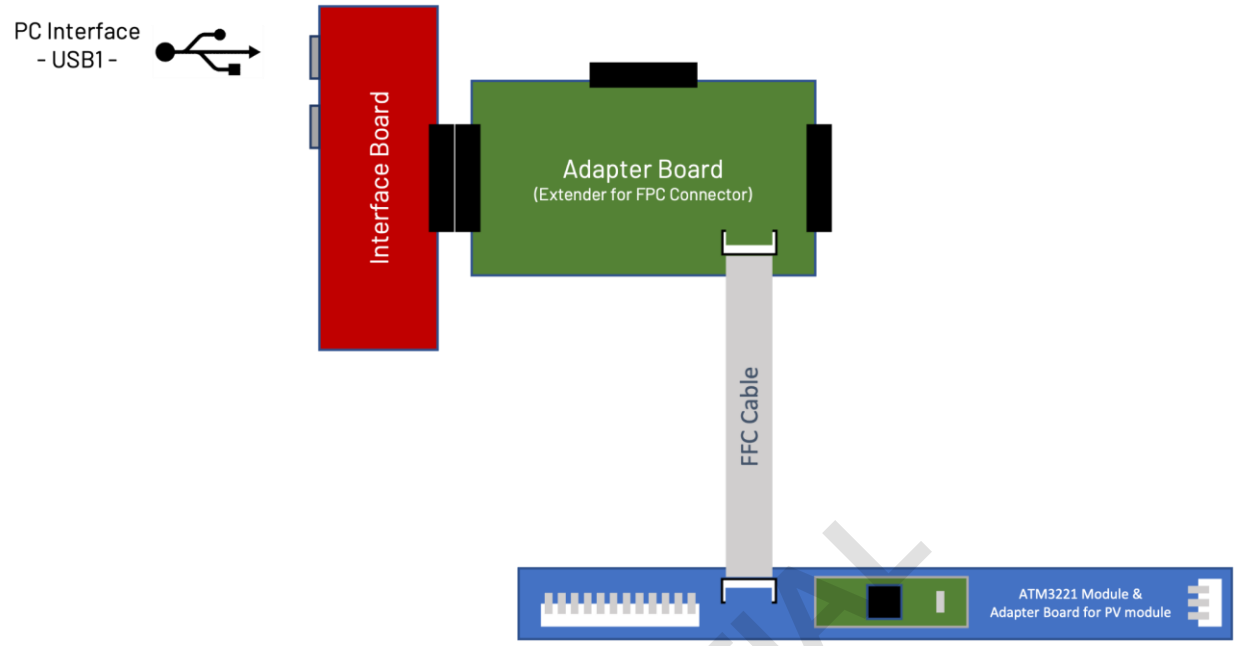


Figure 2 - Adaptor Board with FFC Connector

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2.1 Set-up for Firmware Programming

- Jumpers at the interface board must be installed.
  - P32 (BBOOT)
  - P1 (SWD_CLK)
  - P2 (SWD_IO)
  - PWD
- Jumper at the adapter board (extender board for FFC connector) must be installed.
  - P0 (SWD_EN)
  - TMC
- Connect the interface board to the laptop, then you are ready to program the firmware.

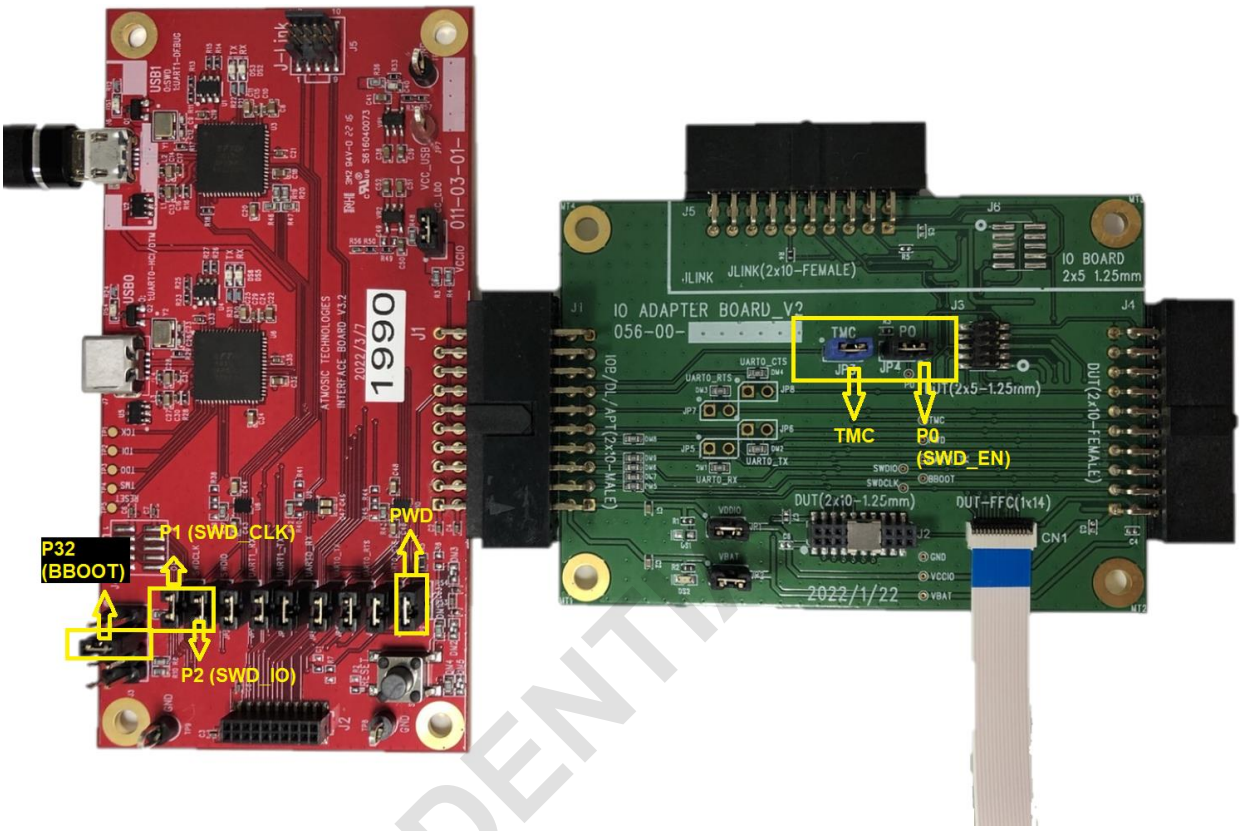


Figure 3 - Interface Board and Adaptor Board Jumper Locations

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2.2 Set-up Keyboard for Normal Operation

2.2.1 Test with console LOG enabled.

- Remove all jumpers of the interface board and Adaptor board (extender board for FFC cable).
- Only keep jumpers for PWD and UART1_TX (P33), see Figure 4.
- Power on the keyboard and you should be able to find the device named “ATM HID KB ”from the laptop, and connect the laptop to the keyboard through Bluetooth.

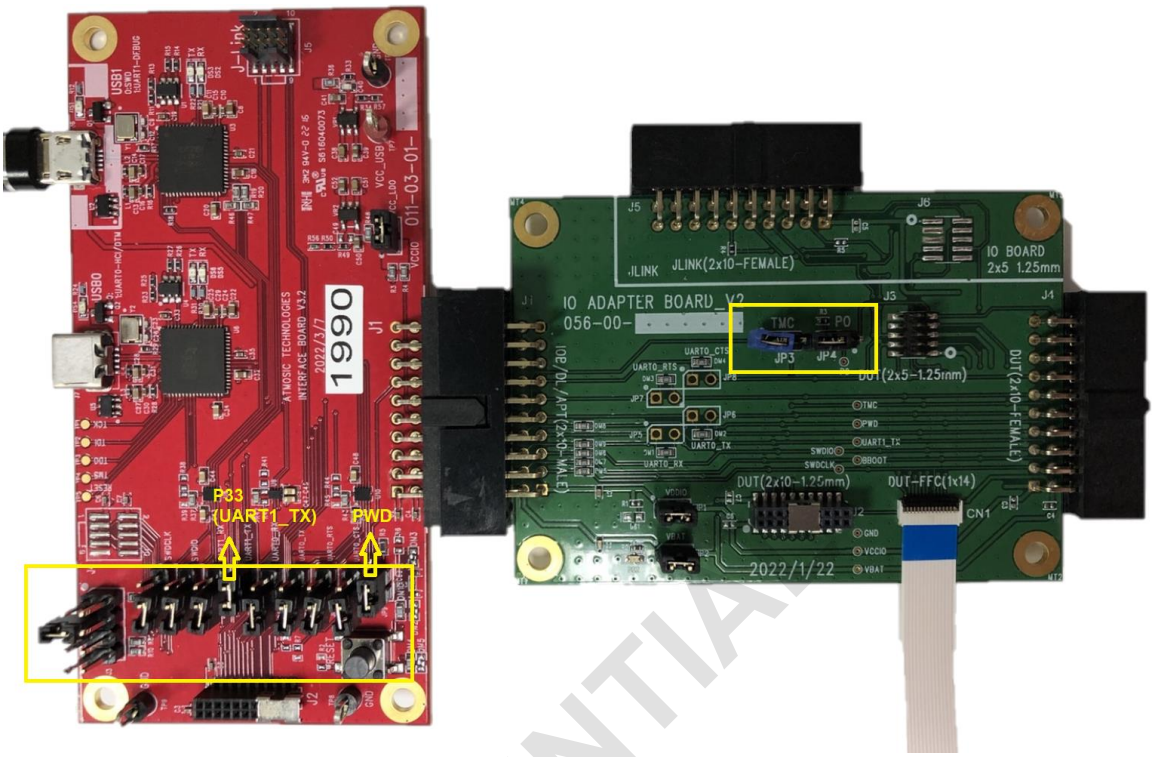


Figure 4 - UART1_TX (P33) and PWD Jumper Locations

2.2.2 Test without console LOG.

- Install the CR2032 battery in the battery housing[1] as shown in Figure 5.
- Connect laptop to keyboard through Bluetooth.

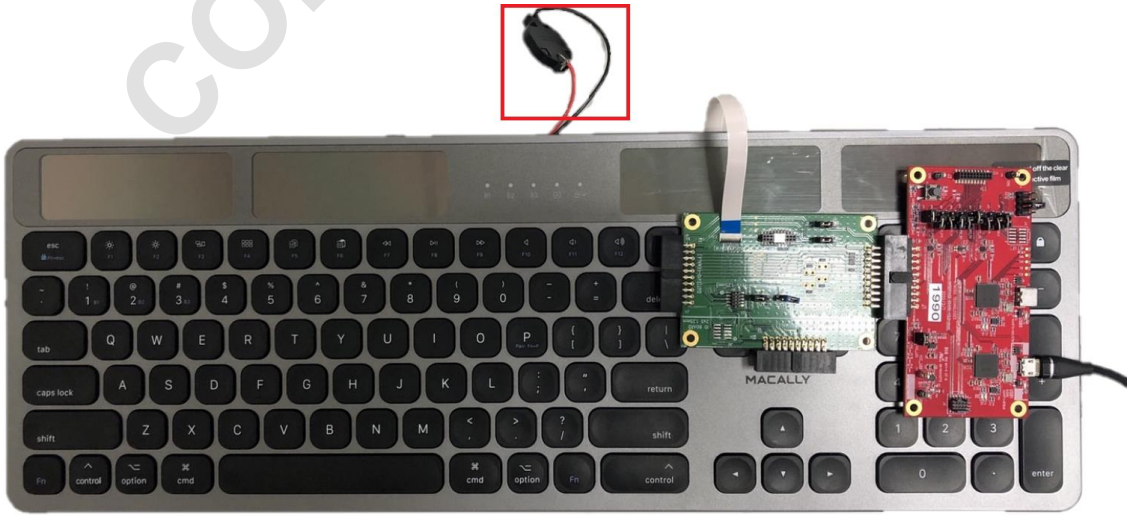


Figure 5 - Connect with Battery

[1] Battery housing: Memory Protective Devices, part number BC2032-E2

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2.3 Set-up Current Consumption Measurement

- Remove the interface board and adaptor board, see Figure 6.
- Connect the power supply as shown in Figure 7, which connects to VBAT of the ATM2/ATM3 device.
- Power on the external power supply, then connect the laptop to the keyboard, and measure the current based on target condition.



Figure 6 - Connect with External Power Supply without Interface Board and Adaptor Board

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Figure 7 - External Power Supply Connections

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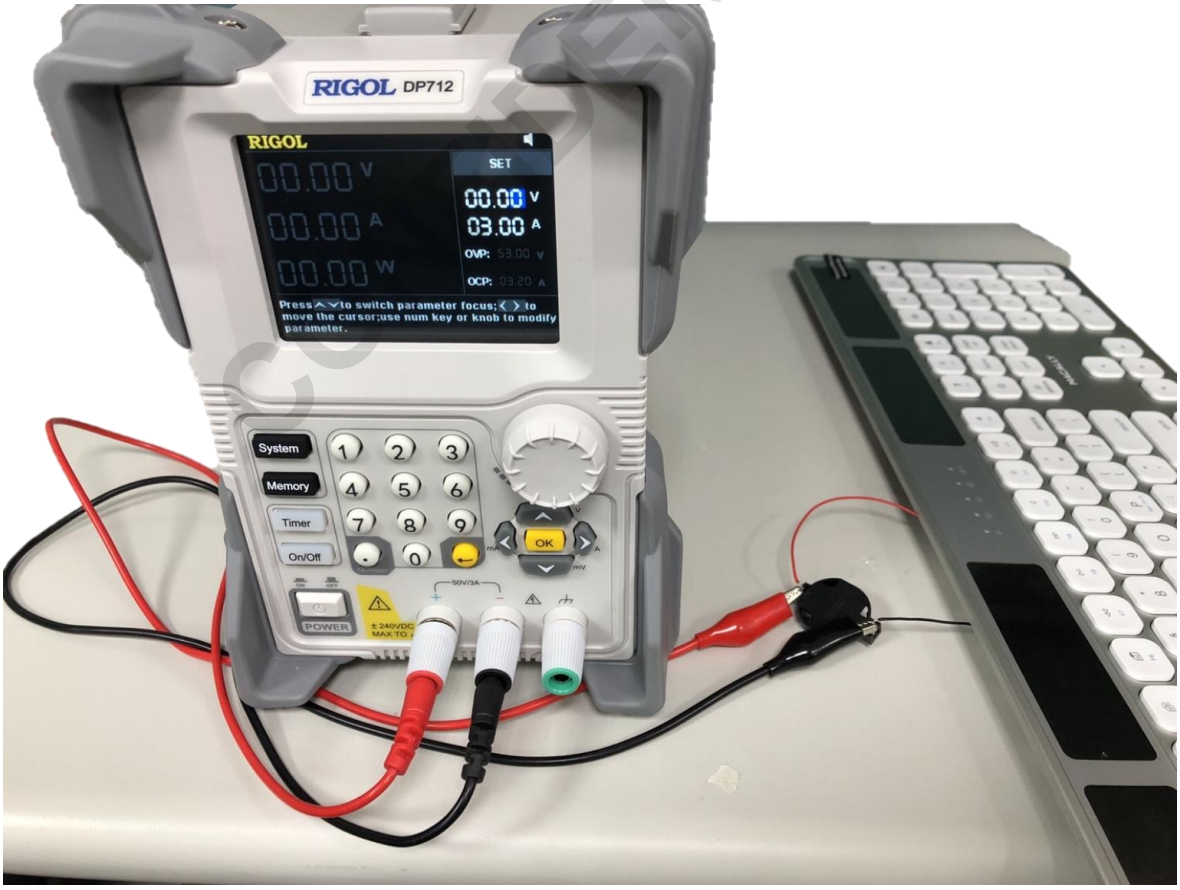


Figure 8 - Current Consumption Measurement Setup

3. Build and Program

3.1 Set up a Command-line Zephyr Development Environment

Follow the Getting Started Guide on Zephyr's website:

Getting Started Guide — Zephyr Project Documentation

3.2 Prepare Zephyr Workspace From Atmosic Github

```
git clone -b wip/keyboard git@github.com:Atmosic/zephyr.git
git clone -b wip/keyboard git@github.com:Atmosic/apps.git
west init -l zephyr
west update cmsis hal_atmosic mbedtls mcuboot tinycrypt
```

3.3 Build and Program Bootloader

- Build bootloader

```
west build -s bootloader/mcuboot/boot/zephyr -b ATMEVK-M2221-02
-d build/ATMEVK-M2221-02/bootloader/mcuboot/boot/zephyr --
-DCONFIG_BOOT_SIGNATURE_TYPE_NONE=y
-DCONFIG_BOOT_SIGNATURE_TYPE_RSA=n
```

- Program bootloader

  ***Note**: ATRDIxxxx, xxxx is the serial number of the interface board*

```
west flash --build-dir
build/ATMEVK-M2221-02/bootloader/mcuboot/boot/zephyr
--device=ATRDIxxxx
```

3.4 Build and Program Application

- Build hid_kb

```
west build -s ./apps/hid_kb -b ATMEVK-M2221-02 -d
build/ATMEVK-M2221-02/zephyr/samples/hid_kb --
-DCONFIG_BOOTLOADER_MCUBOOT=y
-DCONFIG_MCUBOOT_GENERATE_UNSIGNED_IMAGE=y
```

- Program hid_kb

***Note**: ATRDIxxxx, xxxx is the serial number of the interface boar*

```
west flash --build-dir
build/ATMEVK-M2221-02/zephyr/samples/hid_kb --device=ATRDIxxxx
```

3.5 Execute the Code

By default, the log was enabled. After programming the bootloader and application, some messages will be printed to UART1. The developer can connect UART to watch the log. The default baud rate is **115200**. Below is the reference screenshot for first boot.

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```
I: ATM3221-x1x silicon: 6x6 pkg, External flash
I: flash_type:5
*** Booting Zephyr OS build zephyr-v3.0.0-673-g991f06e2bba6  ***
I: Starting bootloader
I: Primary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Scratch: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Boot source: primary slot
I: Swap type: none
I: Bootloader chainload address offset: 0xc000
I: Jumping to the first image slot
*** Booting Zephyr OS build zephyr-v3.0.0-673-g991f06e2bba6  ***
[00:04:27.967,000] <inf> board: ATM3221-x1x silicon: 6x6 pkg, External flash
[00:00:00.002,000] <dbg> kscan_atm: kscan_atm_init: Keyboard HW: 9 ROWS, 17
COLS
[00:00:00.002,000] <dbg> kscan_atm: kscan_atm_init: Keyboard SW: 8 ROWS, 18
COLS
[00:00:00.003,000] <inf> flash_atm: flash_type:5
[00:00:00.009,000] <inf> kbd_mmi: kbd_s_changed last_s:MMI_S_BOOTED
op:MMI_OP_INITING next_s:MMI_S_BOOTED
[00:00:00.010,000] <inf> kbd_mmi: kbd_s_changed last_s:MMI_S_BOOTED
op:MMI_OP_INITING next_s:MMI_S_INITING
[00:00:00.064,000] <inf> fs_nvs: 6 Sectors of 4096 bytes
[00:00:00.065,000] <inf> fs_nvs: alloc wra: 0, dc8
[00:00:00.065,000] <inf> fs_nvs: data wra: 0, 287
[00:00:01.961,000] <inf> bt_hci_core: No ID address. App must call
settings_load()
[00:00:01.962,000] <inf> kbd_gap: Bluetooth initialized
[00:00:01.984,000] <inf> bt_hci_core: Identity[0]: C2:55:66:55:66:55 (public)
[00:00:01.988,000] <inf> bt_hci_core: Identity[1]: C2:55:66:55:66:56 (random)
[00:00:01.992,000] <inf> bt_hci_core: Identity[2]: C2:55:66:55:66:57 (random)
[00:00:01.993,000] <inf> bt_hci_core: HCI: version 5.0 (0x09) revision 0x0013,
manufacturer 0x0060
[00:00:01.994,000] <inf> bt_hci_core: LMP: version 5.0 (0x09) subver 0x0013
[00:00:01.995,000] <inf> kbd_gap: bt id count 3
[00:00:01.996,000] <inf> kbd_gap: kbd_gap_s_changed last_s:GAP_S_INIT
op:GAP_OP_INITED next_s:GAP_S_IDLE
[00:00:01.998,000] <inf> kbd_mmi: kbd_s_changed last_s:MMI_S_INITING
op:MMI_OP_INIT_DONE next_s:MMI_S_IDLE
[00:00:01.999,000] <inf> kbd_gap: [0] Pairing
[00:00:01.999,000] <inf> kbd_gap: Advertising set create...
[00:00:02.027,000] <inf> kbd_gap: Advertising Start
[00:00:02.028,000] <inf> kbd_gap: kbd_gap_s_changed last_s:GAP_S_IDLE
op:GAP_OP_ADV1ING next_s:GAP_S_ADV1ING
[00:00:02.030,000] <inf> kbd_mmi: kbd_s_changed last_s:MMI_S_IDLE
op:MMI_OP_PAIRING next_s:MMI_S_PAIRING
[00:00:02.237,000] <inf> battery: zerovolt channel id 8
```

4. State Machine

State machine is the core of the application. Developers can easily understand the behavior of the device and change/optimize the design. Please refer to MMI State Description and GAP State Description sections. State machine is created by defining the state transition table mechanism provided by the atm_asm module.

4.1 High Level State Description

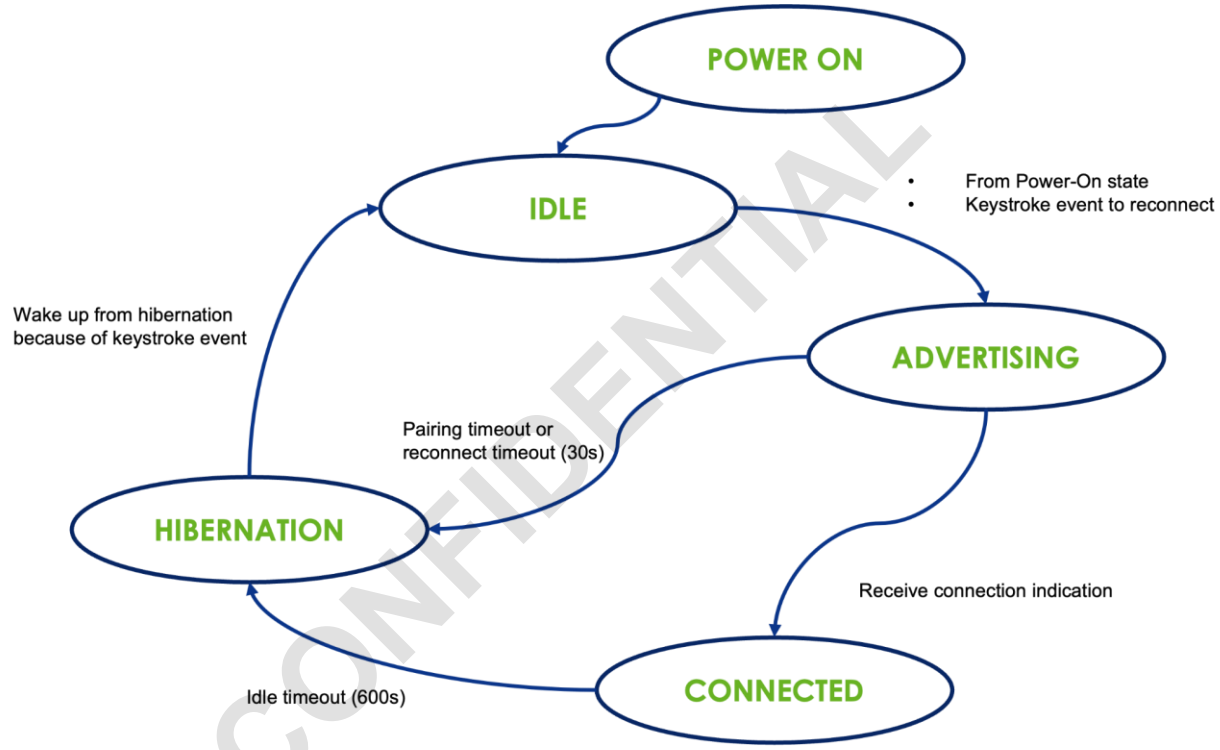


Figure 9 - Top Level State Machine

4.2 MMI State Description

Table 1 - kbd_mmi States

States	Description
MMI_S_BOOTED	This is the initial state, the chip booted from hardware reset, power on reset or hibernate.
MMI_S_INITING	Initialize kbd_ksm, kbd_gap and other modules.
MMI_S_IDLE	Initialization done or enter hibernate. It also acts as an end state.
MMI_S_PAIRING	Bonding with new host.

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States	Description
MMI_S_RECONNING	Reconnecting the bonded host.
MMI_S_CONNECTED	BT linked is connected but HOGP is not ready.
MMI_S_HID_READY	HOGP is ready.
MMI_S_DISCONNING	Keyboard is initiating the disconnect request

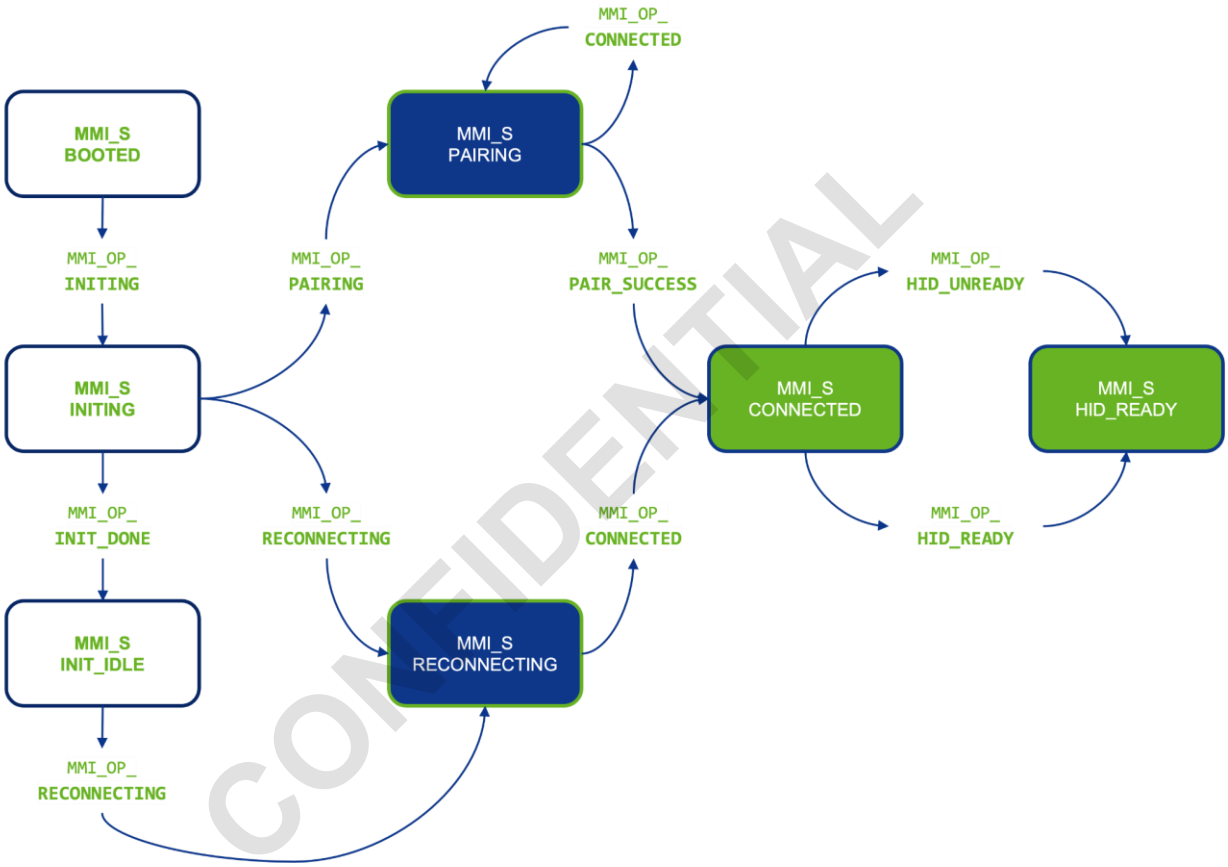


Figure 10 - kbd_mmi State Transitions (Ordinary Cases)

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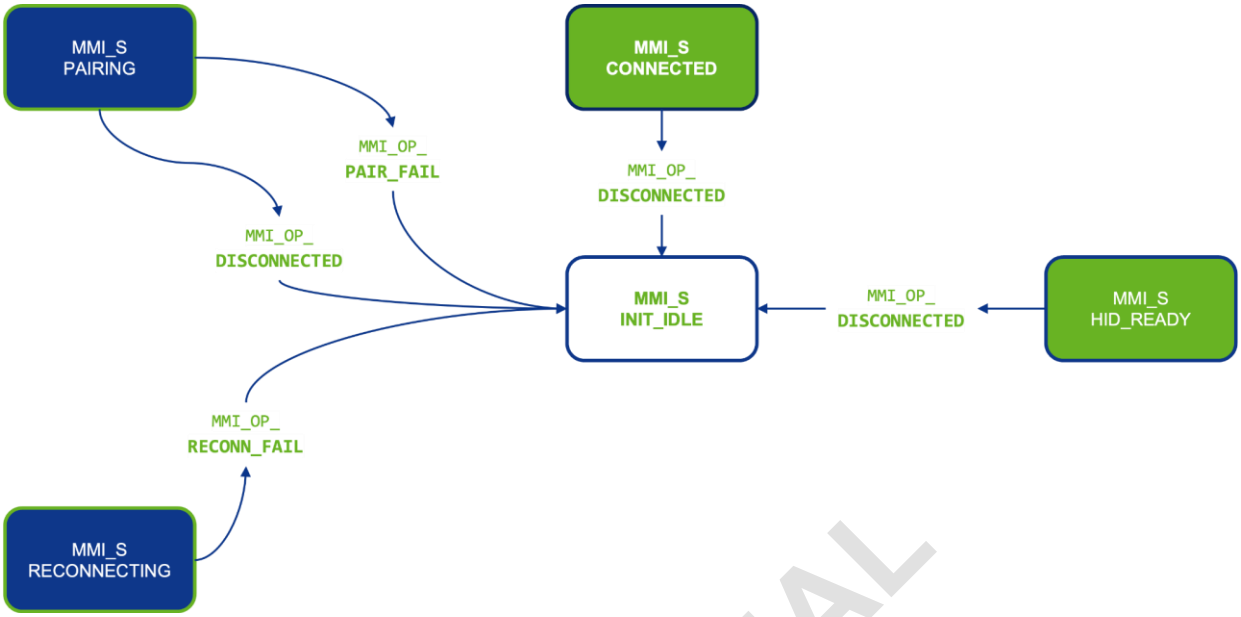


Figure 11 - kbd_mmi State Transitions (Unordinary Cases)

Operations are used to make the state machine transited when something happens. Each operation was triggered by some modules with a certain meaning. Table 2 describes the operations and the triggered module.

Table 2 - kbd_mmi Operations

Operations	Description	Trigger by
MMI_OP_INITING	Initialization is ongoing.	kbd_mmi
MMI_OP_INIT_DONE	Initialization is complete.	kbd_gap
MMI_OP_RECONNING	Device is under advertising for reconnection.	kbd_gap
MMI_OP_PAIRING	Device is under advertising for pairing.	kbd_gap
MMI_OP_RECONN_FAIL	Device is fail to reconnect	kbd_gap
MMI_OP_PAIR_FAIL	Device is fail to pair	kbd_gap
MMI_OP_PAIR_ENTER_KEY_START	Device starts to enter pair key	kbd_gap
MMI_OP_PAIR_ENTER_KEY_DONE	Device end to enter pair key	kbd_gap
MMI_OP_PAIR_SUCCESS	Device is paired successfully.	kbd_gap
MMI_OP_CONNECTED	Device is connected.	kbd_gap
MMI_OP_DISCONNED	Link is disconnected	kbd_gap
MMI_OP_HID_READY	HID is ready to use.	kbd_hog
MMI_OP_HID_UNREADY	HID becomes unready.	kbd_hog

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Operations	Description	Trigger by
MMI_OP_DISCONNING	MMI wants to disconnect the link. It is caused by an expired IDLE timer.	kbd_mmi
MMI_OP_ADV_STOPPED	Advertisement is stopped normally.	kbd_gap

4.3 GAP State Description

As kbd_mmi, the state machine of kbd_gap is created in the application layer. There are seven states and nine operations defined in kbd_gap state machines. Table 3 describes those states, and Figure 11 depicts the state transition.

Table 3 - kbd_gap States

States	Description
GAP_S_INIT	Initial state.
GAP_S_IDLE	No Bluetooth activity is ongoing. Go to hibernate if not further request from kbd_mmi.
GAP_S_ADV0ING	Pairing advertisements is ongoing.
GAP_S_ADV1ING	Reconnecting advertisements is ongoing.
GAP_S_ADV_STOPPING	Advertisement is stopping actively.
GAP_S_CONNECTED	ACL link exists.
GAP_S_RFTEST	DTM mode is ongoing.

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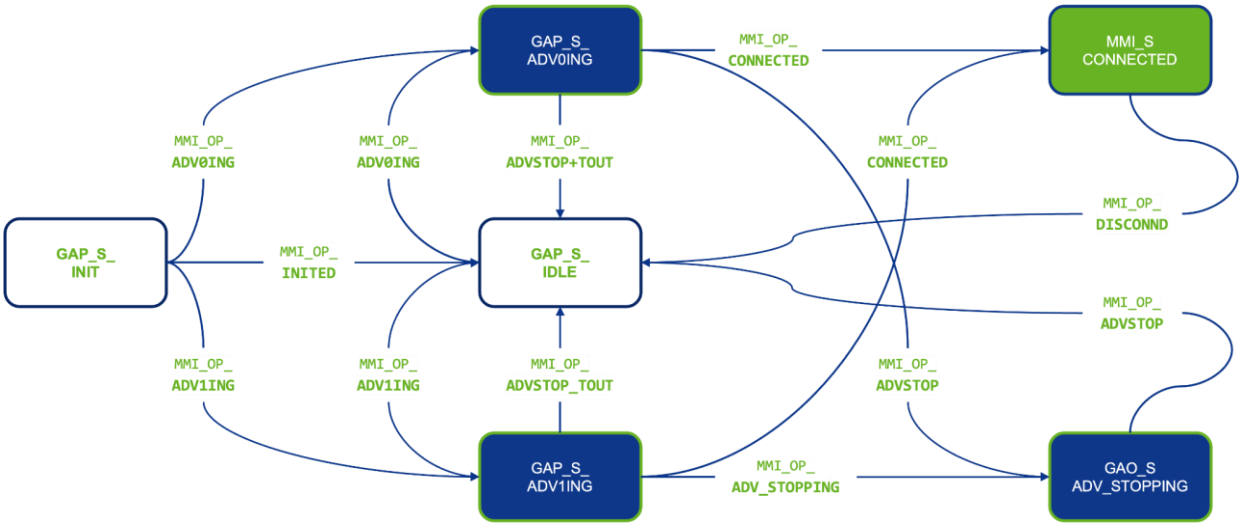


Figure 12 - kbd_gap State Transition

Each operation was triggered by some atm_gap callbacks with a certain meaning. Table 4 describes the operations and the triggering function.

Table 4 - kbd_gap Operations

Operations	Description	Trigger by
GAP_OP_INITING	atm_gap is initialing	kbd_gap_init
GAP_OP_INITED	atm_gap initialization indication	kbd_gap_init_cfm
GAP_OP_ADV0ING	Pairing started confirmation	adv_cfm
GAP_OP_ADV1ING	Reconnecting started confirmation	adv_cfm
GAP_OP_ADV_STOP	ADV stopped by connection	adv_cfm
GAP_OP_ADV_STOP_TOUT	ADV stopped by timeout	adv_cfm
GAP_OP_ADV_STOPPING	ADV Stopping is ongoing by kbd_gap	kbd_gap_stop_discoverable
GAP_OP_CONNECTED	ACL link connection indication	kbd_conn_ind
GAP_OP_DISCONNED	ACL link disconnection indication	kbd_disc_ind

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5. Functional Features

Following functionalities are supported in the Zephyr package

- 8*18 key matrix
- Battery service
- Device information service
- Human interface service
- Advertising
- Connect to the host
- Bond with the host
- Press key and send corresponding keycode to the host via HOGP
- Enter retention mode during connection event
- Idle timeout timer
- Disconnect with the host
- Enter hibernation mode
- Press key to wake up the device from hibernation mode
- Remove all bond (Fn + P)
- Watchdog

6. Limitations and Known Issues

Issue	Power consumption is not optimized
Impact	Recommend to perform power measurements after optimization
Resolution	To be fixed in the next release.

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Revision History

Date	Version	Description
October 10, 2022	0.51	Updated Overview, Hardware Setup, Prepare Zephyr Workspace From Atmosic Github, Build and Program Application,
September 21, 2022	0.50	Initial version created.

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